

Agenda

Short review of single-sample CIs

Confidence intervals

- For the mean of normal random samples with a small sample size and unknown variance
- For the mean difference between two samples
- For the difference between two population proportions

CI - Normal random sample

$X_1, \dots, X_n$: random sample

$X_1, \dots, X_n \sim \text{Normal with mean } \mu \text{ and standard deviation } \sigma.$

Assume that σ is known.

Two-sided CI: A $100(1 - \alpha)\%$ CI for the mean μ of a normal population when σ is known is given by:

$$\left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

One-sided CI: A $100(1 - \alpha)\%$ bounds for the mean μ of a normal population when σ is known is given by:

$$\text{LCB: } \bar{x} - z_{\alpha} \frac{\sigma}{\sqrt{n}} \quad \text{UCB: } \bar{x} + z_{\alpha} \frac{\sigma}{\sqrt{n}}$$

CI – Large random sample

$X_1, \dots, X_n$: random sample

$X_1, \dots, X_n \sim$ with mean μ and standard deviation σ .

Assume that σ is unknown but the sample size is large.

Two-sided CI: A $100(1 - \alpha)\%$ CI for the mean μ of an arbitrary population when n is large is given by:

$$\left(\bar{x} - z_{\alpha/2} \frac{s}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{s}{\sqrt{n}} \right)$$

One-sided CI: A $100(1 - \alpha)\%$ bounds for the mean μ of an arbitrary population when n is large is given by:

$$\text{LCB: } \bar{x} - z_{\alpha} \frac{s}{\sqrt{n}} \quad \text{UCB: } \bar{x} + z_{\alpha} \frac{s}{\sqrt{n}}$$

A Large-Sample CI for a Population Proportion

X : # of successes in a random sample with $n \ll$ population size

p : proportion of 'success' in a population

$$X \sim \text{bin}(n, p) \quad \rightarrow \quad E(X) = np \ \& \ V(X) = np(1 - p)$$

For large n with $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$, X is approximately normal.

A Large-Sample CI for a Population Proportion

$$X \sim \text{bin}(n, p) \quad \rightarrow \quad E(X) = np \text{ \& } V(X) = np(1 - p)$$

For large n with $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$, X is approximately normal.

An approximate confidence interval for a population proportion p with confidence level approximately $100(1-\alpha)\%$ is:

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

New research: An approximate confidence interval for a population proportion p with confidence level approximately $100(1-\alpha)\%$ is:

No need for a
constraint on n and p !

$$\tilde{p} \pm z_{\alpha/2} \sqrt{\frac{\tilde{p}(1 - \tilde{p})}{\tilde{n}}}$$

$$\tilde{n} = n + 4 \text{ and } \tilde{p} = \frac{X+2}{\tilde{n}}$$

Class exercise

In the small city, where the green environment supporters claim that the air is heavily polluted, daily PM10 levels for the last two months (January-February 2018) are recorded.

Stem-and-Leaf Display: PM10

Stem-and-leaf of PM10 N = 59
Leaf Unit = 0.10

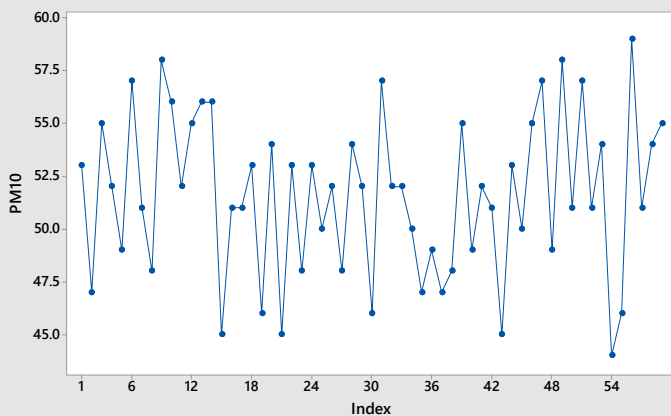
1	44	0
4	45	000
7	46	000
10	47	000
14	48	0000
18	49	0000
21	50	000
28	51	0000000
(7)	52	0000000
24	53	00000
19	54	0000
15	55	00000
10	56	000
7	57	0000
3	58	00
1	59	0

Class exercise

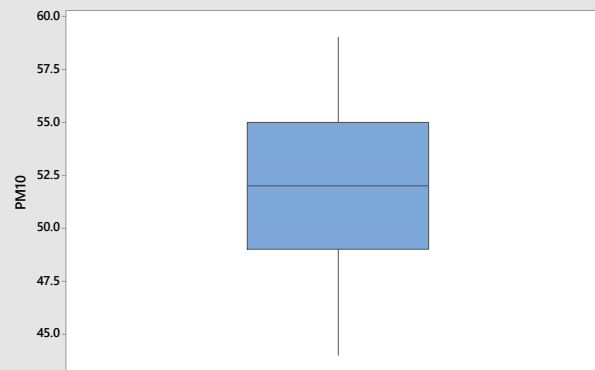
In the small city, where the green environment supporters claim that the air is heavily polluted, daily PM10 levels for the last two months (January-February 2018) are recorded. Using the data below, what can you say about the claim that the average daily PM10 level is more than the upper limit 50mg?

N	Mean	SE Mean	StDev	Min	Q1	Median	Q3	Max	Skewness
59	51.59	0.495	3.802	44	49	52	55	59	-0.10

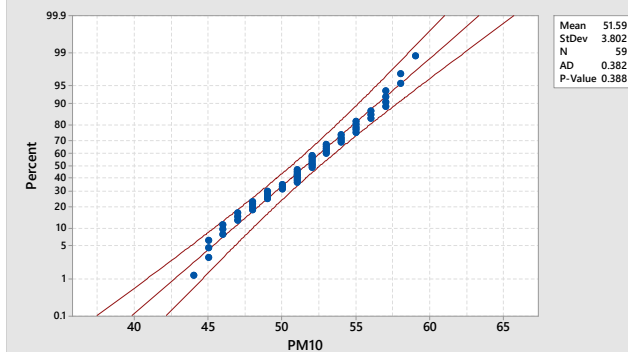
Time Series Plot of PM10



Boxplot of PM10



Probability Plot of PM10
Normal - 95% CI



Can we find confidence levels or bounds?

In the small city, where the green environment supporters claim that the air is heavily polluted, daily PM10 levels for the last two months (January-February 2018) are recorded.

The PM10 levels cause extreme hazard if the daily level is strictly greater than 55ppm. Green environment supporters claim that the proportion of extremely hazardous days is strictly positive. Evaluate this claim using appropriate confidence intervals or bounds.

Stem-and-Leaf Display: PM10

Stem-and-leaf of PM10 N = 59

1	44	0
4	45	000
7	46	000
10	47	000
14	48	0000
18	49	0000
21	50	000
28	51	0000000
(7)	52	0000000
24	53	00000
19	54	0000
15	55	00000
10	56	000
7	57	0000
3	58	00
1	59	0

$$n = 59 \text{ \& } \hat{p} = \frac{10}{59}$$

$$\tilde{n} = 63 \text{ \& } \tilde{p} = \frac{12}{63}$$

Computations

Confidence Intervals for the Mean of a Normal r.v. with Unknown Variance:

$X_1, X_2, \dots, X_n$: random sample with small n

Assumptions:

- $X_i \sim N(\mu, \sigma)$
- both μ and σ are unknown.

The change in the distribution of the random variable

The random variable under consideration is still:

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

How about its distribution? Still a $N(0,1)$ r.v.?

NO!!! Why?

When n is small, S is not necessarily close to σ , thus T has more variability than Z , or in other words, T will be more spread out than the standard normal distribution.

Definition: t Distribution

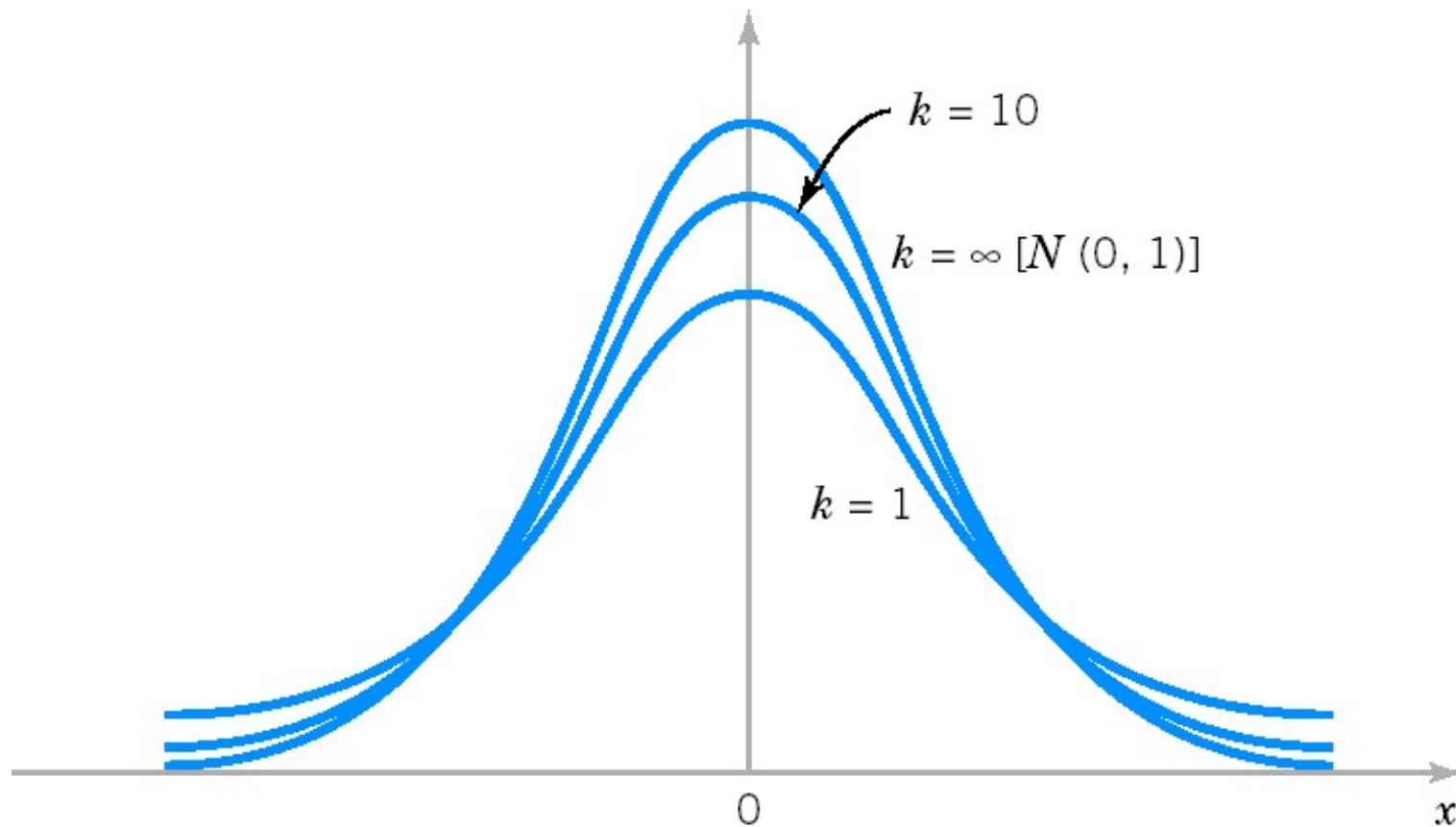
Let ν be a positive integer. Then a random variable X is said to have a t distribution with parameter ν , and the pdf of X is given by:

$$f(x; \nu) = \frac{\Gamma((\nu + 1)/2)}{\sqrt{\pi\nu}\Gamma(\nu/2)} \frac{1}{((x^2/\nu) + 1)^{(\nu+1)/2}} \quad -\infty < x < \infty.$$

The parameter ν is called the number of degrees of freedom (df) of X .

Difficult to compute, so the corresponding values are tabulated (so tables are provided)

The t distribution

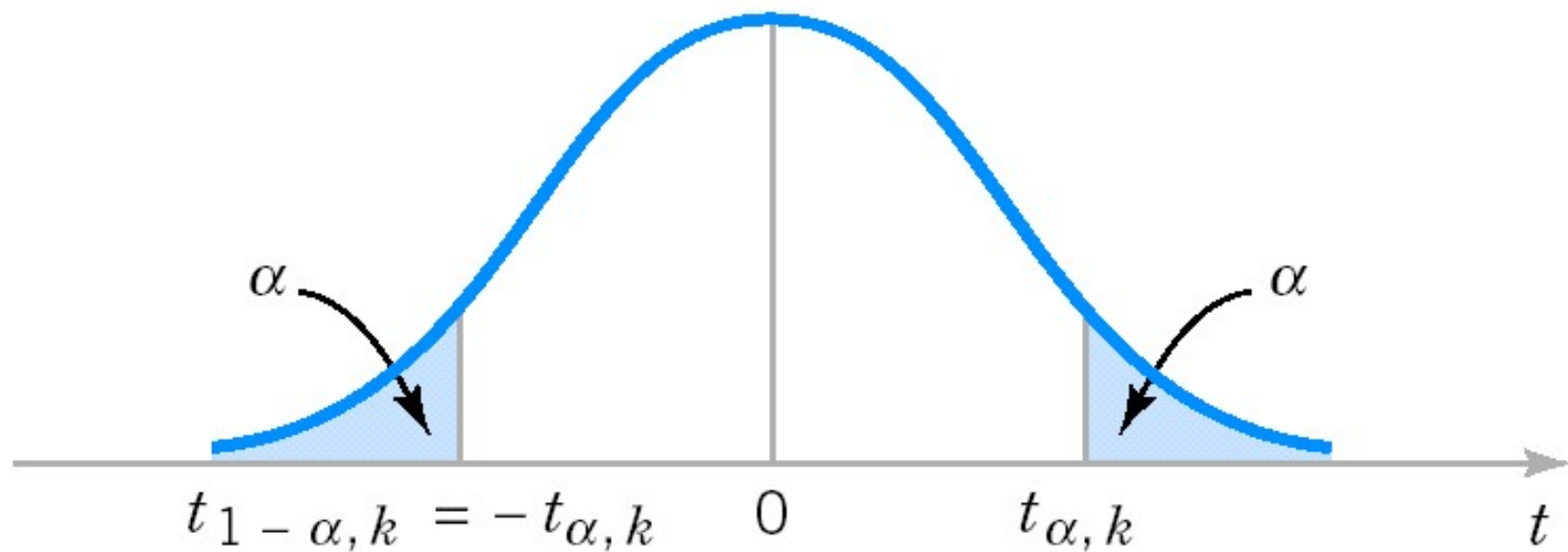


Probability density functions of several t distributions.

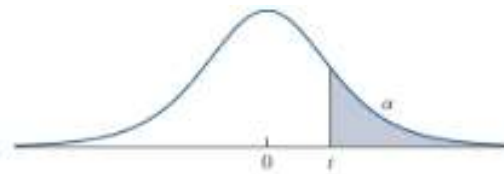
Notation

$t_{\alpha, v}$ is the critical value

$t_{\alpha, v}$ is decreasing in both α & v .



t-table



v	α								
	0.40	0.25	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
1	0.325	1.000	3.078	6.314	12.706	31.821	63.657	318.309	636.619
2	0.289	0.816	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.277	0.765	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.271	0.741	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.267	0.727	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.265	0.718	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.263	0.711	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.262	0.706	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.261	0.703	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.260	0.700	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.260	0.697	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.259	0.695	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.259	0.694	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.258	0.692	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.258	0.691	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.258	0.690	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.257	0.689	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.257	0.688	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.257	0.688	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.257	0.687	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.257	0.686	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.256	0.686	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.256	0.685	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.256	0.685	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.256	0.684	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.256	0.684	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.256	0.684	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.256	0.683	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.256	0.683	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.256	0.683	1.310	1.697	2.042	2.457	2.750	3.385	3.646
35	0.255	0.682	1.306	1.690	2.030	2.438	2.724	3.340	3.591
40	0.255	0.681	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.254	0.679	1.296	1.671	2.000	2.390	2.660	3.232	3.460
120	0.254	0.677	1.289	1.658	1.980	2.358	2.617	3.160	3.373
∞	0.253	0.674	1.282	1.645	1.960	2.326	2.576	3.090	3.291

Can we find a CI around μ ?

The random variable under consideration is still:

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

How about its distribution? Still a $N(0,1)$ r.v.?

NO!!! Why?

When n is small, S is not necessarily close to σ , thus T has more variability than Z , or in other words, T will be more spread out than the standard normal distribution.

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

Proposition

Let $\bar{x}$ and s be the sample mean and sample standard deviation computed from the results of a random sample of n from a normal population with mean μ . Then,

$$\left(\bar{x} - t_{n-1, \alpha/2} \frac{s}{\sqrt{n}}, \bar{x} + t_{n-1, \alpha/2} \frac{s}{\sqrt{n}} \right)$$

is a confidence interval for μ with confidence level $100(1-\alpha)\%$.

Example: A cargo company

A cargo company aims to have a mean delivery time of 5 hours for local deliveries.

They believe that delivery times are normal with σ is not known.

They recorded 10 delivery times as:

7.3, 3.3, 4.7, 6.8, 10.1, 5.7, 6.4, 4.2, 3.1, 8.9

95%CI on the true mean delivery time?

95% CI with known $\sigma = 2.4$:
(4.56, 7.54), with width = 2.98

$$\bar{x} = 6.05$$

$$s = 2.32$$

95% CI with unknown σ estimated by $s = 2.32$:
(4.39, 7.71), with width = 3.32

How to derive a CI?

Main idea:

- Collect a random sample
- Determine the parameter of interest, say θ
- Find a random variable, say Y , that involves parameter θ and the random sample.
- Specify the distribution of Y .
- Use the distribution of Y to find random lower and upper bounds on θ

Parameter of interest: true mean, μ

- $X_1, \dots, X_n$: A random sample with $E[X] = \mu$
- We are looking for a random variable $Y = h(X_1, \dots, X_n; \mu)$ whose distribution can be specified.

	X is normal		X is not normal		
	$n < 40$	$n \geq 40$	$n < 30$	$30 \leq n < 40$	$n \geq 40$
σ : known	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$	?	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$
σ : unknown	$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$	$Z = \frac{\bar{X} - \mu}{S/\sqrt{n}}$	?	?	$Z = \frac{\bar{X} - \mu}{S/\sqrt{n}}$

CIs around the true mean, μ

- $X_1, \dots, X_n$: A random sample with $E[X] = \mu$
- We are looking for a random variable $Y=h(X_1, \dots, X_n; \mu)$ whose distribution can be specified.

	X is normal		X is not normal		
	$n < 40$	$n \geq 40$	$n < 30$	$30 \leq n < 40$	$n \geq 40$
σ : known	$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$	$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$	?	$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$	$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$
σ : unknown	$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$	$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$	?	?	$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$

Parameter of interest: true population proportion p

X : Number of successes in n trials, each trial is i.i.d. with constant $p = P\{\text{success}\}$

We are looking for a random variable, $Y = h(X; \theta)$ whose distribution can be specified.

$$\hat{p} = \frac{X}{n}$$

$$n\hat{p} \geq 10 \text{ and } n(1 - \hat{p}) \geq 10$$

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}}$$

CIs around the true population proportion p

X : Number of successes in n trials, each trial is i.i.d. with constant $p = P\{\text{success}\}$

We are looking for a random variable, $Y = h(X; \theta)$ whose distribution can be specified.

$$\hat{p} = \frac{X}{n}$$

$$\tilde{p} = \frac{X + 2}{n + 4}$$

$$\tilde{n} = n + 4$$

$$n\hat{p} \geq 10 \text{ and } n(1 - \hat{p}) \geq 10$$

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

No condition on n or p

$$\tilde{p} \pm z_{\alpha/2} \sqrt{\frac{\tilde{p}(1 - \tilde{p})}{\tilde{n}}}$$

Comparing two populations

Men versus Women

Adults versus Children

Traditional Treatment versus New Treatment

...

Inferences based on two samples

$X_1, \dots, X_{n_X}$: a random sample from a population with μ_X & σ_X^2

$Y_1, \dots, Y_{n_Y}$: a random sample from a population with μ_Y & σ_Y^2

X & Y samples are independent of one another.

Can we find a confidence interval around $\mu_X - \mu_Y$?

Estimate $\mu_X - \mu_Y$?

What can you say about $\bar{X} - \bar{Y}$? $E(\bar{X} - \bar{Y}) = ?$
 $V(\bar{X} - \bar{Y}) = ?$

Can we find random variable with a known distribution?

The random variables that may work are:

$$\frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}} \quad \& \quad \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}}$$

Their distributions?

Normal samples & known variances

$X_1, \dots, X_{n_X}$: a random sample from a normal population with μ_X & σ_X^2

$Y_1, \dots, Y_{n_Y}$: a random sample from a normal population with μ_Y & σ_Y^2

X & Y samples are independent of one another.

When the values of σ_X and σ_Y are known, a $100(1 - \alpha)\%$ CI for $\mu_X - \mu_Y$ is:

$$\bar{X} - \bar{Y} \pm z_{\alpha/2} \sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}.$$

Arbitrary distributions, unknown variances, large sample sizes

$X_1, \dots, X_{n_X}$: a random sample from a population with μ_1 & σ_X^2

$Y_1, \dots, Y_{n_Y}$: a random sample from a population with μ_2 & σ_Y^2

X & Y samples are independent of one another.

When the two samples are from arbitrary distributions, and the values of σ_X and σ_Y are unknown, a $100(1 - \alpha)\%$ CI for $\mu_X - \mu_Y$ is:

$$\bar{X} - \bar{Y} \pm z_{\alpha/2} \sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}.$$

Normal samples, unknown variances, small sample sizes

$X_1, \dots, X_{n_X}$: a random sample from a population with μ_X & σ_X^2

$Y_1, \dots, Y_{n_Y}$: a random sample from a population with μ_Y & σ_Y^2

X & Y samples are independent of one another.

What is the distribution of the following?

$$\frac{\bar{X} - \bar{Y} - \mu_X - \mu_Y}{\sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}} \sim t_v$$

$$v = \left[\frac{\left(\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y} \right)^2}{\frac{\left(\frac{s_X^2}{n_X} \right)^2}{n_X - 1} + \frac{\left(\frac{s_Y^2}{n_Y} \right)^2}{n_Y - 1}} \right]$$

Normal samples, unknown variances, small sample sizes

100(1 - α)% CI for the difference $\mu_X - \mu_Y$ is

$$\bar{X} - \bar{Y} \pm t_{v, \alpha/2} \sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}.$$

$$v = \left\lceil \frac{\left(\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y} \right)^2}{\frac{\left(\frac{s_X^2}{n_X} \right)^2}{n_X - 1} + \frac{\left(\frac{s_Y^2}{n_Y} \right)^2}{n_Y - 1}} \right\rceil$$

When the Populations Have Equal Variances

Suppose we have the same set-up as before, but the populations are known to have nearly the same variance.

Then a $100(1 - \alpha)\%$ CI for $\mu_X - \mu_Y$ is

$$\bar{X} - \bar{Y} \pm t_{n_X+n_Y-2, \alpha/2} \cdot s_p \sqrt{\frac{1}{n_X} + \frac{1}{n_Y}}.$$

The quantity s_p is the pooled standard deviation, given by

$$s_p = \sqrt{\frac{(n_X - 1)s_X^2 + (n_Y - 1)s_Y^2}{n_X + n_Y - 2}}.$$

The best practice is to assume the variances are unequal unless it is quite certain that they are equal.